

Saksham Panghal

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EDUCATION

Indian Institute of Technology, Jodhpur

Bachelor of Technology, Computer Science & Engineering

CGPA: **8.55**

Aug. 2024 – May 2028

Central Board of Secondary Education

Senior Secondary Examination

Percentage: **93.6%**

Mar. 2024

Central Board of Secondary Education

Secondary School Examination

Percentage: **99.0%**

Mar. 2022

EXPERIENCE

Undergraduate Researcher at SPADE Lab

IIT Jodhpur

Jan. 2026 – Apr. 2026

Jodhpur, Rajasthan

- Benchmarked and enhanced 5 Graph Neural Network architectures (GCN, GraphSAGE, GAT, GIN, ChebNet) using **PyTorch Geometric** to improve connected-component membership classification across real-world graph datasets with up to **875K nodes** and **7M+ edges** and synthetic graphs with up to 100K nodes.
- Improved Macro-F1 to **92.1%** through focal loss based **imbalance handling** and **adaptive sampling**, achieving a **22.7%** gain over baseline methods on large scale real-world graph datasets.
- Developed an automated benchmarking framework evaluating Accuracy, Macro-F1, and Balanced Accuracy across **transductive** and **inductive** settings, enabling systematic comparison across multiple graph datasets.

PROJECTS

Collaborative Code Workspace | Node.js, Yjs, PostgreSQL, Socket.io, Docker |

Apr. 2026 – Jun. 2026

- Architected a real-time, multi-user collaborative coding workspace using React, Node.js, **Socket.IO**, and **Yjs**, enabling **low-latency distributed state synchronization** with **CRDT** based eventual consistency.
- Designed a scalable backend architecture combining **in-memory state caching** with **PostgreSQL persistence**, minimizing database overhead while providing fault-tolerant recovery for collaborative sessions.
- Implemented **concurrent** room lifecycle management, user presence tracking, backend services, and **REST APIs** to support scalable real-time collaboration under concurrent user workloads.

College Utility Platform | Flutter, Firebase, FastAPI, Python, Dart |

Feb. 2026 – Apr. 2026

- Developed a **modular** cross-platform Flutter application integrating **6** backend services for ride sharing, lost & found, messaging, notifications, and user management using Firebase and FastAPI.
- Designed a **real-time ride matching engine** using a dual centroid geofencing algorithm with constraint-aware pooling (2 km origin, 5 km destination, ± 30 minute windows), maintaining $O(N)$ scalability under simulated workloads of up to **1 million concurrent ride requests**.
- Integrated an AI-driven microservice utilizing **Pinecone** vector databases for **semantic image matching** in the lost and found module, achieving **94% accuracy** in item identification.

Metro Route Planner | Node.js, React, PostgreSQL, Redis, Docker |

Mar. 2026 – May 2026

- Engineered a high throughput transit routing system over a **GTFS**-derived network with **262** stations, **36** routes, and **128K+ stop-time** records to compute shortest time, minimum transfer, and fare-optimized paths.
- Ingested and normalized GTFS transit feeds into a weighted graph of **262 stations** and **548+ station connections**, implementing **Dijkstra's Algorithm** and **BFS** for efficient route computation and transfer optimization.
- Reduced average route-query latency by over **75% (about 4,500 ms)** through graph preprocessing and a **Redis**-backed caching layer for frequently queried routes.

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, JavaScript, HTML/CSS, Dart

Frameworks & Libraries: React, Node.js, Next.js, FastAPI, Tailwind CSS, Socket.IO, Yjs, NetworkX, PyTorch Geometric

Database & Services: PostgreSQL, MySQL, MongoDB, Firebase, Redis, Prisma ORM

Developer Tools & Platforms: Git, Docker, Vercel, Linux, Postman

Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Distributed Systems, Concurrency, Synchronization, REST APIs, Caching, Graph Neural Networks (GNNs), Conflict-Free Replicated Data Types (CRDTs)

RELEVANT COURSEWORK

Data Structures and Algorithms, Mathematics for Computing, Software Engineering, Pattern Recognition and Machine Learning, Introduction to Computer Science, Probability, Statistics & Stochastic Processes

ACHIEVEMENTS

- Co-authoring a research manuscript on connectivity-aware Graph Neural Networks (under preparation for publication), resulting from research at SPADE Lab, IIT Jodhpur.
- Achieved an All India Rank of **2480** in JEE Advanced 2024
- Achieved **99.81 percentile** in JEE Main 2024, securing a place in the **top 0.19%** of **1.4M+** candidates.